Capacitor Charge and Energy 1

Essential Pre-Uni Physics I1.1

Subject & topics: Physics | Electricity | Capacitors **Stage & difficulty:** A Level P2

Complete the questions in the table.

Capacitance	Voltage / V	Charge / C	Energy / J
$100\,\mu\mathrm{F}$	6.0	(a)	(b)

Part A

Charge

a) What is the charge stored in the capacitor?

Part B

Energy

b) What is the energy stored in the capacitor?

Question deck:

STEM SMART Physics 32 - Capacitors

Capacitor Charge and Energy 5

Essential Pre-Uni Physics I1.5

Subject & topics: Physics | Electricity | Capacitors **Stage & difficulty:** A Level P2

Calculate the capacitance of a capacitor needed in a back up power supply if it needs to store $0.24\,\mathrm{J}$ of electrical energy when connected to a $12\,\mathrm{V}$ power supply.

Question deck:

STEM SMART Physics 32 - Capacitors

Capacitor Charge and Energy 8

Essential Pre-Uni Physics I1.8

Subject & topics: Physics | Electricity | Capacitors **Stage & difficulty:** A Level P2

A mystery capacitor can store $3.0\,\mathrm{J}$ of energy when connected to a $10\,\mathrm{V}$ supply. How much energy can it store when connected to a $5.0\,\mathrm{V}$ supply?

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Capacitor Networks 3

Essential Pre-Uni Physics I2.3

Subject & topics: Physics | Electricity | Capacitors **Stage & difficulty:** A Level P2

A $200\ \mu\text{F}$ capacitor is in series with a $2200\ \mu\text{F}$ capacitor. The capacitors are charged until the $200\ \mu\text{F}$ capacitance has a voltage of $12\ \text{V}$ across it. What is the voltage across the $2200\ \mu\text{F}$ capacitor?

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Capacitor Networks 4

Essential Pre-Uni Physics I2.4

Subject & topics: Physics | Electricity | Capacitors **Stage & difficulty:** A Level C2

A $470\ \mu\mathrm{F}$ capacitor is charged using a $10\ \mathrm{V}$ battery. It is then disconnected from the battery, and connected to an uncharged $220\ \mu\mathrm{F}$ capacitor. Calculate the voltage across the capacitors once the current has stopped flowing. (Hint: capacitors are effectively in parallel, and total charge has not changed.)

Question deck:

STEM SMART Physics 32 - Capacitors

Photoelectric Capacitor

100 Challenge Problems in Pre-Uni Physics Q54

Subject & topics: Physics | Electricity | Capacitors **Stage & difficulty:** A Level C1

An uncharged, isolated parallel plate capacitor has capacitance C. Monochromatic light of frequency f is incident on one of the metal plates, which has threshold frequency f_0 where $f > f_0$. Emitted photoelectrons are then captured by the other plate and eventually the capacitor charge reaches a steady value.

Part A

Number of electrons

Find an expression for the number of electrons transferred between the plates in terms of C, f, f_0 and other constants.

(Use h for Planck's constant and e for the magnitude of the electron charge.)

The following symbols may be useful: `C, e, f, f_0, h`

Part B

Fraction of energy stored

Assuming each transferred electron is emitted by a single photon, find an expression for the fraction of the total energy of these photons that is stored in the capacitor.

The following symbols may be useful: `f, f_0`

Created for the British Physics Olympiad and isaacscience.org by Kimlam Nguyen

An Oscillating Capacitor

Subject & topics: Physics | Electricity | Capacitors **Stage & difficulty:** A Level C2

A capacitor with capacitance C is attached to an AC voltage source, which gives a voltage $V = V_0 \sin \omega t$.

If there is no resistance in the circuit, $C = 330\,\mu\text{F}$, $V_0 = 12.0\,\text{V}$, and $\omega = 5720\,\text{rad}\,\text{s}^{-1}$, what is the magnitude of the current flowing at $t = 3.42\,\text{s}$?

Created for isaacphysics.org by Michael Conterio

Question deck:
STEM SMART Physics 32 - Capacitors